**VQE for the half-filled Hubbard ring on MIMIQ: procedure and first results**

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# 1 Goal

Prepare the ground state of the Hubbard ring with the variational quantum eigensolver (VQE) [1] on the MIMIQ simulator [2], then compute its $n$BRDMs and GME and compare them with exact diagonalization (ED). System: $L = 6$ sites, $K = 6$ electrons (half filling), $S_z = 0$, $V = 0$, $t = 1$; this needs 12 qubits. Code, tests and results: `https://github.com/ItsKhuramShahzad/mimiq-gme`.

| Symbol | Meaning |
|---|---|
| $E_0^{\mathrm{ED}},\ \lvert\Psi_0^{\mathrm{ED}}\rangle$ | exact ground-state energy and state, from exact diagonalization (ED) |
| $E_1^{\mathrm{ED}}$ | exact energy of the first excited state (ED) |
| $\Delta = E_1^{\mathrm{ED}} - E_0^{\mathrm{ED}}$ | energy gap |
| $E_{\mathrm{VQE}},\ \lvert\Psi(\boldsymbol{\theta}^*)\rangle$ | lowest energy found by VQE and its state |
| $E_{\mathrm{HF}}$ | energy of the Hartree–Fock starting state |

# 2 Model

$$H_e = -t\sum_{k=1}^{L}\sum_{\sigma}\left(c_{k,\sigma}^{\dagger}c_{k+1,\sigma} + c_{k+1,\sigma}^{\dagger}c_{k,\sigma}\right) + U\sum_{k=1}^{L} n_{k,\uparrow}\, n_{k,\downarrow} + V\sum_{k=1}^{L} n_k\, n_{k+1} + \epsilon\sum_{k=1}^{L} n_k, \tag{1}$$

with $n_{k,\sigma} = c_{k,\sigma}^{\dagger}c_{k,\sigma}$, $n_k = n_{k,\uparrow} + n_{k,\downarrow}$ and site $L+1 \equiv 1$ (ring). At fixed $K$ the $\epsilon$ term only adds the constant $\epsilon K$, so $\epsilon = 0$ is used. Since $H_e(t, U, V) = t\, H_e(1, U/t, V/t)$, the ground state depends only on $U/t$ and $V/t$.

This Hamiltonian is never changed. VQE does not diagonalize it; it only measures its expectation value.

# 3 From electrons to qubits (Jordan–Wigner)

One qubit per spin orbital; $|1\rangle$ = occupied, $|0\rangle$ = empty:

$$q(k,\sigma) = 2(k-1) + \sigma, \qquad \sigma = 0\ (\uparrow),\ 1\ (\downarrow), \tag{2}$$

so the 12 qubits are $1\uparrow, 1\downarrow, 2\uparrow, 2\downarrow, \ldots, 6\downarrow$. The Jordan–Wigner transformation [3] is

$$c_{k,\sigma} = Z_0 \cdots Z_{q-1}\,\tfrac{1}{2}(X_q + iY_q), \qquad n_{k,\sigma} = \tfrac{1}{2}(1 - Z_q). \tag{3}$$

The $Z$ string gives the fermion sign. The terms of Eq. (1) become ($q < q'$)

$$c_q^{\dagger}c_{q'} + c_{q'}^{\dagger}c_q = \tfrac{1}{2}\left(X_qZ_{q+1}\cdots Z_{q'-1}X_{q'} + Y_qZ_{q+1}\cdots Z_{q'-1}Y_{q'}\right), \tag{4}$$

$$n_q\, n_{q'} = \tfrac{1}{4}\left(1 - Z_q - Z_{q'} + Z_qZ_{q'}\right). \tag{5}$$

So the Hamiltonian is a sum of Pauli strings with real coefficients,

$$H_e = \lambda_0 + \sum_\mu \lambda_\mu P_\mu, \tag{6}$$

with 42 strings for $L = 6$ (24 hopping, 18 on-site) and $\lambda_0 = UL/4$. This step is exact: the spectrum does not change.

## 4 How VQE works

A circuit with parameters $\boldsymbol{\theta}$ (the *ansatz*) prepares a trial state $|\Psi(\boldsymbol{\theta})\rangle$. MIMIQ measures

$$E(\boldsymbol{\theta}) = \langle\Psi(\boldsymbol{\theta})|H_e|\Psi(\boldsymbol{\theta})\rangle = \lambda_0 + \sum_\mu \lambda_\mu \langle P_\mu\rangle_{\boldsymbol{\theta}} \;\geq\; E_0^{\mathrm{ED}}. \tag{7}$$

A classical optimizer (COBYLA [4], which needs no gradients) changes $\boldsymbol{\theta}$ to lower $E(\boldsymbol{\theta})$, and the loop repeats until the energy stops decreasing. By the variational principle the lowest energy found is an upper bound to $E_0^{\mathrm{ED}}$, and the state approaches $|\Psi_0^{\mathrm{ED}}\rangle$.

## 5 Initial state

The circuit starts from the ground state at $U = 0$, which is easy to prepare.

*Orbitals.* Diagonalize the $L \times L$ one-electron hopping matrix $h_{kk'} = -t(\delta_{k',k+1} + \delta_{k',k-1})$:

$$h = C\,\mathrm{diag}(e_1, \ldots, e_L)\,C^T, \qquad c_{k,\sigma} = \sum_p C_{kp}\, a_{p,\sigma}, \qquad e_p = -2t, -t, -t, t, t, 2t. \tag{8}$$

The orbitals are only used to build the start; the energy is always measured with the full $H_e$.

*Hartree–Fock.* Fill the three lowest orbitals with both spins:

$$|\Phi_0\rangle = \prod_{p=1}^{3} a^\dagger_{p,\uparrow} a^\dagger_{p,\downarrow} |\mathrm{vac}\rangle = X_0 X_1 \cdots X_5 |0\cdots 0\rangle, \qquad E_{\mathrm{HF}} = -8t + \tfrac{3}{2}U. \tag{9}$$

This is a closed shell for $L = 6$.

*Back to the sites.* The circuit $\mathcal{R}_C$ rewrites the state from orbitals to sites,

$$\mathcal{R}_C\, a^\dagger_{p,\sigma}\, \mathcal{R}_C^\dagger = \sum_k C_{kp}\, c^\dagger_{k,\sigma}. \tag{10}$$

It is a product of 15 Givens rotations of neighboring orbitals [5]; each one is exactly two Pauli rotations per spin, so there is no approximation. The state $\mathcal{R}_C|\Phi_0\rangle$ is the exact $U = 0$ ground state on the sites. Its overlap with the true ground state is $F = 0.980$, $0.689$, $0.265$ at $U/t = 1, 4, 10$: a good start at small $U/t$, a poor one at large $U/t$.

## 6 Ansätze

*UCCSD* [6] (117 parameters). Single and double excitations from occupied to empty orbitals, $|\Psi\rangle = \mathcal{R}_C\, e^{T(\boldsymbol{\theta}) - T(\boldsymbol{\theta})^\dagger} |\Phi_0\rangle$, applied with one first-order Trotter step.

*LUCJ* [7] (130 parameters). Local unitary cluster Jastrow: orbital rotations and density–density phases, two repetitions.

*HVA* (Hamiltonian variational ansatz [8, 9], $3p$ parameters). Split $H_e$ into three groups whose terms commute: the on-site term $H_U = \sum_k n_{k\uparrow} n_{k\downarrow}$, the hopping on the bonds $(1,2)$, $(3,4)$, $(5,6)$ ($H_A$), and the hopping on the bonds $(2,3)$, $(4,5)$, $(6,1)$ ($H_B$). Then

$$|\Psi(\boldsymbol{\theta})\rangle = \prod_{l=1}^{p} e^{-i\theta_B^{(l)} H_B}\, e^{-i\theta_A^{(l)} H_A}\, e^{-i\theta_U^{(l)} H_U}\, \mathcal{R}_C |\Phi_0\rangle. \tag{11}$$

Inside each group all Pauli strings commute, so every layer is exact (no Trotter error). Every layer conserves $K$, $S_z$ and $S^2$, so the state stays a singlet. At $\boldsymbol{\theta} = 0$ it is the $U = 0$ ground state.

*Warm start along $U/t$.* The HVA circuit does not depend on $U$, and the ground state changes smoothly with $U/t$ (the gap stays open). Therefore the points $U/t = 1, 2, 3, 4, 5, 6, 8, 10$ are solved in this order, each one starting from the optimal angles of the previous one, $\boldsymbol{\theta}_{\text{start}}(U_{j+1}) = \boldsymbol{\theta}^*(U_j)$.

## 7 Accuracy criterion

Two quantities are reported, the energy error and the fidelity (the fidelity is available on the simulator):

$$\Delta E = E_{\text{VQE}} - E_0^{\text{ED}}, \qquad F = |\langle \Psi_0^{\text{ED}} | \Psi(\boldsymbol{\theta}^*) \rangle|^2. \tag{12}$$

They are linked by the gap $\Delta = E_1^{\text{ED}} - E_0^{\text{ED}}$:

$$F \;\geq\; 1 - \frac{\Delta E}{\Delta}. \tag{13}$$

At large $U/t$ the gap is small ($\Delta = 1.66$ at $U/t = 1$, $0.27$ at $U/t = 10$), so a small $\Delta E$ alone does not guarantee a good state. The proposed criterion is $\Delta E < 0.01\,\Delta$ (which gives $F \geq 0.99$), together with $|\Delta \mathcal{C}_{\text{GME}}| < 0.01$.

## 8 Results ($L = 6$, $t = 1$, $V = 0$)

| $U/t$ | ansatz | parameters | $\Delta E$ | $F$ |
|---|---|---|---|---|
| 1 | UCCSD | 117 | $1.6 \times 10^{-4}$ | 0.99997 |
| | LUCJ | 130 | $4.9 \times 10^{-3}$ | 0.9990 |
| | HVA, $p = 8$ | 24 | $2.8 \times 10^{-4}$ | 0.99994 |
| 10 | UCCSD | 117 | $3.9 \times 10^{-1}$ | 0.942 |
| | LUCJ | 130 | $2.1 \times 10^{-1}$ | 0.634 |
| | HVA, $p = 8$, start $\boldsymbol{\theta} = 0$ | 24 | $6.6 \times 10^{-2}$ | 0.952 |
| | HVA, $p = 8$, warm start | 24 | $3.8 \times 10^{-3}$ | **0.99937** |

HVA with $p = 8$ and the warm start along $U/t$:

| $U/t$ | $\Delta$ | $\Delta E$ | $\Delta E/\Delta$ | $F$ |
|---|---|---|---|---|
| 1 | 1.656 | $2.4 \times 10^{-3}$ | 0.15 % | 0.99954 |
| 2 | 1.313 | $2.1 \times 10^{-3}$ | 0.16 % | 0.99957 |
| 3 | 1.009 | $3.8 \times 10^{-3}$ | 0.37 % | 0.99922 |
| 4 | 0.770 | $3.9 \times 10^{-3}$ | 0.51 % | 0.99929 |
| 5 | 0.601 | $4.1 \times 10^{-3}$ | 0.68 % | 0.99929 |
| 6 | 0.485 | $3.3 \times 10^{-3}$ | 0.67 % | 0.99942 |
| 8 | 0.349 | $3.6 \times 10^{-3}$ | 1.03 % | 0.99944 |
| 10 | 0.274 | $3.8 \times 10^{-3}$ | 1.39 % | 0.99937 |

Main observations:

- At small $U/t$ every ansatz works.
- At large $U/t$ a lower energy does not mean a better state: LUCJ has a lower energy than UCCSD at $U/t = 10$ but a much lower fidelity, because it mixes in low-lying spin excitations that cost little energy.
- HVA with $p = 8$ and the warm start gives $F \geq 0.9992$ from $U/t = 1$ to 10 with only 24 parameters. The criterion $\Delta E < 0.01\,\Delta$ holds up to $U/t = 6$ and is just missed at $U/t = 8$ and 10, where the gap is small.
- A test of entanglement: the entanglement entropy of 1, 2 and 3 sites of the HVA states agrees with ED within 0.02 bits.

# 9 Next steps

1. $n$BRDMs of the HVA states measured from Pauli strings on MIMIQ, GME with the GCG solver, compared with ED at every $U/t$.
2. A cheaper circuit for the digital twin [10]: ordering the qubits as $1\uparrow, \ldots, 6\uparrow, 1\downarrow, \ldots, 6\downarrow$ makes every hop (except across the ring bond) a two-qubit gate, which halves the number of two-qubit gates of the $p = 8$ circuit ($1616 \to 856$), with exactly the same state.

# References


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